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Does the loss function need to match the data-generating distribution for regression tasks?

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Abstract

Every model fitted by optimisation requires a loss function, and for regression the choice is usually made between squared error and absolute error on grounds of robustness to outliers. Each of the common regression losses is, however, the negative log-likelihood of a distribution, so choosing a loss means assuming a distribution for the response — the same assumption that in a generalised linear model fixes the link, the variance function and the inference. Classical theory predicts that the matching assumption should be the most efficient, but whether this holds for neural networks, which make no commitment about the form of the mean, has rarely been tested. I simulated Gaussian- and Poisson-distributed data with known effects and fitted one correctly specified GLM alongside five networks differing only in their loss, over 1000 iterations per scenario, comparing predictive accuracy, effect recovery and interval coverage. On right-skewed count data the Poisson and negative binomial losses reduced error against the true conditional mean by 9% and 8% relative to squared error. On Gaussian data, where all five were expected to perform alike, the two count losses were again ahead, by 12% and 20%. Mean absolute error failed outright: it estimates the conditional median rather than the mean, predicted below zero for 13% of count observations, and scored worse than an intercept-only baseline. These differences were visible only against the true mean; against the observed response, the only comparison available on real data, they fell below 1.5%. Effect recovery did not follow the same ordering: every network underestimated the strongest Poisson effect by 13–23%, with coverage as low as 0.70, and the matching loss was not the least biased. The correctly specified GLM outperformed every network throughout. Matching the loss to the response distribution therefore improves estimation of the conditional mean but not inference, and a loss targeting the wrong functional can fail badly on sparse counts — a risk absent from current guidance on loss selection.

1. Introduction

Machine learning now shapes work across science, industry and everyday life. Applying it means making a series of choices: which algorithm suits the problem, how to prepare the data, which hyperparameters to tune. In ecology, most applications are supervised and fall into either classification or regression, depending on whether the response is categorical or numeric (Pichler & Hartig, 2023). For regression, one choice receives far less attention than the others, although it defines what the model is optimising in the first place: the loss function. Any model fitted by numerical optimisation requires one, and the losses in routine use are not arbitrary. Mean squared error, mean absolute error and the Poisson and negative binomial losses can each be derived as the negative log-likelihood of a corresponding distribution, so that specifying a distribution for the response determines the loss (Goodfellow et al., 2016). Choosing a loss therefore means assuming a distribution for the response, whether or not the choice is made deliberately. In classical statistics, that assumption has both a principled justification and a long-running argument about whether it matters in practice.

Fisher (1922) gave the first general justification for fitting models by maximising the likelihood: of all the ways of reducing a sample to a few numbers, the maximum-likelihood estimate retains the whole of the information the sample carries about the parameter, so that, asymptotically, no competing estimator is more precise. He treated the choice of distribution, however, as a problem separate from estimation and an explicitly empirical one, in which the mathematical form of the population is chosen from experience and tested against the data afterwards. Wedderburn (1974) narrowed how much of that form is actually required, showing that estimation needs only a specified relationship between the mean and the variance rather than a complete distribution. White (1982) established what fails when the assumption is wrong: the estimator converges to a well-defined limit that may still be the quantity of interest, while the standard errors and tests derived from the assumed likelihood are not valid. Gneiting (2011) added that any such loss is consistent for one particular property of the predictive distribution — squared error for the mean, absolute error for the median — so that evaluating a model under a loss aimed at a different quantity can be severely misleading. So the distributional assumption buys efficiency when it holds, and what fails first when it does not is the inference rather than the estimate itself. Whether that advantage is large enough to matter in practice has been contested in ecology for the past fifteen years.

In ecology, count data have traditionally been handled in one of two ways: transforming the response and fitting a linear model (LM), or fitting a generalised linear model (GLM) whose family matches the assumed distribution of the data. O'Hara & Kotze (2010) argued for the latter. Transformation aims to stabilise the variance and linearise the response, and usually requires adding a constant to avoid log(0), but in their simulations the transformed LMs were substantially biased, while quasi-Poisson and negative binomial GLMs recovered the mean with little bias. Ives (2015) reached the opposite conclusion for a different question. When testing whether a regression coefficient differs from zero, LMs on transformed data held nominal type I error rates, whereas GLMs produced false positives even when correctly specified — although once this was corrected, the GLMs retained slightly greater power. Warton et al. (2016) argued that these are separable problems: the model should be chosen to match the properties of the data, and poor type I error control should be fixed afterwards through the testing procedure, for instance by resampling. They also showed that better-fitting models can have considerably more power. Most recently, Morrissey & Ruxton (2020) demonstrated that much of O'Hara and Kotze's result followed from comparing two different quantities, since the LM estimated E[log(y + 1)] while the GLM estimated log(E[y]); once both were expressed on the same scale, an LM fitted to the untransformed counts performed indistinguishably from the correctly specified GLM. A practical cost of the identity link nonetheless persists in both studies: LMs can return negative fitted values for counts (O'Hara & Kotze, 2010; Warton et al., 2016). Fifteen years on, the requirement that a model's distribution match the data is therefore neither established nor refuted, and much of the disagreement turns on which quantity is being estimated or tested. In the last decade, a third option to handle count data in ecology has emerged: The Deep Neural Network (DNN).

What changes when looking at a DNN's loss function from the perspective of a statistician is which parts of the classical specification survive. In a GLM, the distribution family does three things at once. It sets the link, constraining the scale on which predictors act and the range the fitted mean can take; it sets the variance function, determining how strongly each observation is weighted during fitting; and it supplies the likelihood from which standard errors and tests are derived (McCullagh & Nelder, 1989; Ver Hoef & Boveng, 2007). A neural network keeps only some of this. Because networks are universal function approximators, they make no commitment to a functional form for the mean (Goodfellow et al., 2016; LeCun et al., 2015), and they provide no analytic standard errors, so uncertainty must be obtained by resampling.

The loss function, however, still carries the other two jobs. The output activation fixes the link, and with it whether predictions can leave the range of the response: an exponential output cannot return a negative count, an identity output can. And the loss still weights observations: squared error treats a given deviation as equally serious wherever it occurs, absolute error weights it differently again, while a Poisson loss encodes Var = μ and so penalises the same absolute deviation far more severely at small counts than at large ones.

In practice these two components receive little attention. Model development in deep learning focuses overwhelmingly on architecture and hyperparameter tuning, with the loss left at its default — typically squared error, or absolute error where outliers are a concern (Elharrouss et al., 2025; Lathuilière et al., 2020; Terven et al., 2025). Where loss choice is discussed for regression, the argument is usually about robustness to outliers rather than about the distribution of the response, so that the mean–variance relationship and the link, which determine the family in a GLM, play little part. The few direct tests come from genomic prediction. Montesinos-López et al. (2021) trained a Poisson-loss network on count data and compared it with an otherwise identical network trained under squared error; the Poisson loss did not outperform it, and a companion study found that generalised Poisson regression outperformed the corresponding networks (Montesinos-López et al., 2020). Both studies, however, assessed predictions against observed counts rather than against the underlying mean, and in count data the irreducible sampling variance can dominate that comparison, leaving little scope to resolve a difference in estimation efficiency (Oliveira et al., 2025). Neither examined effect recovery, and neither extended beyond the Poisson–Gaussian contrast.

This study addresses that gap. Does a loss function matching the data-generating distribution give the best predictive accuracy and the best recovery of effects? I simulate data from two known distributions, Gaussian and Poisson, and fit one correctly specified GLM alongside five neural networks differing only in their loss function, comparing predictive accuracy against the true conditional mean as well as against the observed response, and comparing estimated effects against the values used to generate the data. Following the classical argument, the matching loss should be the most efficient, so the Poisson and negative binomial losses should perform best on Poisson data and all five should perform similarly on Gaussian data; following the ecological literature, any such advantage may be too small to matter. The question has a practical motivation: residual diagnostics are the standard tool for model criticism in classical statistics, and a comparable rule for choosing a loss would be useful for neural networks. That

the matching loss is in fact the best one is a precondition for any such rule, and it is that precondition which is tested here.

2. Materials and Methods

Data Creation

Data were simulated with DHARMa::createData, which generates a response variable (observedResponse) from three predictors (Environment1, Environment2, Environment3) under a specified distribution. I ran two scenarios with different distributions, Gaussian and Poisson, at 1000 iterations each. Effect sizes were set to 2, 0.4 and 0.1 respectively to test a strong, a moderate and a small effect, so that accuracy and inference could be assessed across effect magnitudes. Overdispersion and random-effect variance were set to zero to keep the structure as simple as possible, although varying them alongside intercept and effect sizes would be a natural extension.

The intercept differed between scenarios. For Poisson data I set it to −1; a low mean makes count data strongly right-skewed, where the mismatch between count and continuous losses should be most pronounced. The same intercept proved unusable for Gaussian data, where roughly two thirds of the simulated responses were negative, and the Poisson and negative binomial losses failed to converge (Table S1, see appendix). I therefore raised the Gaussian intercept to 6, the lowest value that reliably produced only positive responses, keeping the mean-to-variance ratio as small as possible while allowing all five losses to be fitted. The residual standard deviation was left at the default of 1 (scale = 1), so the Gaussian responses had a mean of 6 and a standard deviation of approximately 1.

Sample size was set to 250 so that, after an 80/20 train–test split and a further 20% validation split within training, each network was fitted on 160 observations. This keeps the sample small, since predictive performance degrades and becomes more variable at low sample sizes (Wisz et al., 2008), while still leaving enough data for the networks to converge.

GLM-fit

As a baseline, I fitted a GLM with the family matching the data-generating distribution. To make the comparison with the DNNs, the GLM was also bootstrapped: 20 resamples of the

training data were drawn with replacement, a GLM refitted on each, and predictions averaged across them. Effect estimates were taken as the mean of the 20 bootstrap estimates, standard errors as their standard deviation, and p-values from a z-test. Predictions were generated for both training and test data, from which RMSE, Spearman correlation and R^2 were calculated.

DNN-fit

Five DNNs were fitted per iteration, differing only in their loss function. MSE, MAE and Gaussian represent commonly used losses with an identity link; Poisson and negative binomial represent losses with a log link, chosen because count data are common regression problems in ecology.

Each network had two hidden layers of 50 nodes, allowing some flexibility while restricting complexity. Networks were trained for up to 200 epochs at a learning rate of 0.003. Rather than tuning the learning rate separately for each combination of loss function and data distribution, I used early stopping with a patience of 20 epochs, so that training halts once the validation loss stops improving. This keeps the optimisation settings identical across losses, at the cost of not optimising any of them individually. Weight decay of 0.01 penalised large weights and so mitigated overfitting; I checked for residual overfitting by comparing accuracy metrics on training and test data (Figure S1 and S2,conver see appendix). For each iteration and model, 20 bootstrap models were trained to extract Standard Error (SE) and p-values.

Two diagnostic variables were recorded alongside the accuracy metrics. Neg_preds counts predictions below zero, which are impossible for count data and indicate a loss function predicting outside the support of the response. Each fit was additionally wrapped in a tryCatch, so that a failed fit was recorded as a missing row rather than terminating the iteration, allowing convergence failures to be reported as a rate per loss function.

Accuracy metrics

Four metrics were calculated per model and iteration, on both training and test predictions.

RMSE measures the deviation between predicted and observed responses on the original scale, so smaller values indicate predictions closer to the observed data. It is defined:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{1}$$

RMSE against the true mean is treated as the primary measure of predictive accuracy, because it isolates estimation error from the irreducible sampling variation that dominates the error against the observed response in sparse count data. Because the data were simulated, the noise-free mean μ is known, and RMSE can also be calculated against it rather than against the noisy observations:

$$RMSE_{true} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\mu_i - \hat{y}_i)^2} \tag{2}$$

R^2, the coefficient of determination, compares a model's squared prediction error against that of an intercept-only baseline and so measures whether a model predicts better than the mean. This allows for judging performance without reference to the scale of the data:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3}$$

Because R^2 is built from squared errors, observations with large values dominate it; for right-skewed count data a small number of high counts determines the metric almost entirely. For the Poisson scenario I therefore used McFadden's pseudo-R^2, which replaces sums of squares with log-likelihoods and so evaluates predictions under the assumed count distribution rather than on the squared-error scale:

$$R^2_{McF} = 1 - \frac{\ln L(\hat{M})}{\ln L(M_0)} \tag{4}$$

Since models trained with MSE, MAE or Gaussian losses do not themselves define a count distribution, their predictions were treated as Poisson means for this evaluation. This amounts

to scoring every model with the logarithmic score, a strictly proper scoring rule (Gneiting 2011), normalised against a common baseline. Note that neither metric is bounded below by zero out of sample: negative values indicate predictions worse than the baseline.

Because McFadden's R^2 is computed from Poisson log-likelihoods via dpois(), which is undefined for a negative rate, predictions were clamped to a lower bound of 1e-8. The clamp is unavoidable rather than a modelling choice: no finite Poisson log-likelihood exists at a non-positive rate, so without it the metric would return NaN for any model producing negative predictions. It is, however, a severe penalty. The Poisson log-density contains a term in $\log(\lambda)$, so a clamped prediction paired with a non-zero count yields a very large negative contribution, and a model whose predictions fall below zero for a substantial share of observations is therefore scored far below the intercept-only baseline. The sign of such an R^2 is robust, but its magnitude depends on the chosen bound and should not be interpreted quantitatively; the Neg_preds count is reported alongside it as a bound-independent measure of the same behaviour.

Spearman correlation measures predictive performance independently of scale and bias. Because it uses only ranks, it distinguishes a model that has learned the correct pattern but predicts it on the wrong scale from one that has not learned the pattern at all.

Inference metrics

Unlike a GLM, a DNN has no coefficients to extract, so effects were compared via Average Conditional Effects (ACE), obtained from cito's fitted model object (Pichler & Hartig, 2023). For comparability the equivalent quantity was extracted from the GLM as Average Marginal Effects using marginaleffects::avg_slopes. ACE and AME are the same metric under different names, and both are reported on the response scale; I refer to them as ACE from here on.

Standard errors were obtained from the 20 bootstrap networks per model as the standard deviation of their estimates, and p-values from a z-test of estimate over standard error (Amesöder et al., 2024). Effects were evaluated on the training data, where the true effect is defined.

Estimates were compared against the true average effect implied by the data-generating process. Because this quantity depends on the link, it differs between scenarios: under the

identity link it equals the simulated coefficient, whereas under the log link it equals the coefficient multiplied by the mean response, and therefore varies between iterations. Each estimate was compared against the true value for its own iteration.

Across the 1000 iterations I report bias, the mean difference between estimate and truth, and coverage, the proportion of iterations in which the interval estimate ± 1.96 SE contained the true effect. Coverage tests whether a model's self-reported uncertainty is honest: a well-calibrated model should achieve approximately 95%. Monte Carlo standard errors are reported for both, following (Morris et al., 2019); at 1000 iterations the Monte Carlo error on coverage is approximately 0.007, so differences smaller than roughly 0.02 should not be interpreted.

Two caveats apply to the intervals. The cito documentation notes that standard errors obtained under regularisation tend to be conservative, so the intervals reported here are likely wider than they would be without weight decay. In addition, intervals were formed as estimate ± 1.96 SE from 20 bootstrap replicates; with so few replicates a normal quantile is optimistic, and the corresponding t quantile would be 2.09. This alone implies a coverage of about 0.935 rather than 0.95 even for a perfectly calibrated model, and accounts for most of the shortfall observed for the GLM.

Model comparison

All six models were fitted to the same simulated dataset within an iteration, so their results are paired rather than independent. Differences between models were therefore assessed within iterations, which removes the between-iteration variation and makes small but systematic differences detectable.

For each metric I report the mean across iterations together with its Monte Carlo standard error, SD/√1000, which quantifies how precisely 1000 iterations determine the expected performance rather than the spread between iterations.

Whether the six models differed at all was tested with a repeated-measures ANOVA per metric, with iteration as a blocking factor, and summarised by partial η^2, the share of within-iteration variance attributable to the loss function. Because MAE behaved qualitatively differently from the other losses, each ANOVA was also run on the remaining five models. With 1000 iterations the associated p-values are uninformative, so η^2 is reported instead.

Differences between individual models were quantified as paired differences from the MSE loss, computed within each iteration and summarised by their mean, a 95% confidence interval and Cohen's dz. MSE was chosen as the reference because it is the default loss for regression in practice, so the contrasts answer directly whether changing the loss improves on the default; the GLM appears as one further contrast. P-values were adjusted with the Holm correction across the five contrasts within each metric.

Parallelisation

Each script ran 1000 iterations, each fitting one GLM and five DNNs with 20 bootstrap replicates. Runtime was therefore substantial, and iterations were distributed across 12 cores using pbmcapply::pbmclapply. Results from each iteration were appended to a data frame and written to disk on completion. Analyses were run on a machine with 24 cores and took approximately 28 hours per scenario.

3. Results

All trained models converged on Gaussian- and Poisson-distributed data, an earlier Gaussian configuration in which two losses failed entirely is described below. I performed an overfitting analysis to validate the DNNs' results (see Figures S1 and S2 in the appendix).

Poisson-distributed data

All models converged in every iteration. Predictive accuracy differed systematically between loss functions: the repeated-measures ANOVA gave partial η^2 of 0.52 for McFadden's R^2 and 0.51 for RMSE against the true mean, so the choice of loss accounted for roughly half the within-iteration variance in both. For the rank-based Spearman correlation the share was much smaller ($\eta^2 = 0.14$), as expected for a metric that is insensitive to scale. Excluding MAE, which behaved qualitatively differently from the other losses, η^2 fell to 0.14 for McFadden's R^2 but remained at 0.46 for RMSE against the true mean, so on the primary accuracy measure the differences among the remaining four losses were substantial in their own right rather than driven by a single poorly performing loss.

No network matched the correctly specified GLM on any metric (Table 1). Against the true mean, the GLM achieved an RMSE of 0.156 (SE 0.002) compared with 0.223 for the best net-

work, so the best network carried 43% more error than the parametric model fitted to the same data.

Among the networks the two count losses performed best. Relative to the MSE loss, the Poisson loss reduced RMSE against the true mean by 0.023, a reduction of 9%, and the negative binomial loss by 0.019; both confidence intervals excluded zero (Figure 1). The Gaussian loss was indistinguishable from MSE ($p = 0.45$, difference 0.001), which is consistent with the two differing only in whether the residual standard deviation is estimated. The same ordering held for McFadden's R^2 (Poisson +0.044, negative binomial +0.043, Gaussian +0.011 relative to MSE) and for Spearman correlation, where all differences were below 0.011.

The size of these differences depended strongly on the scale of evaluation. Measured against the observed response rather than the true mean, the advantage of the Poisson loss over MSE fell to 0.8% (0.846 against 0.853). The two quantities are related by $\mathrm{RMSE}^2(\text{observed}) = \sigma^2 + \mathrm{RMSE}^2(\text{true mean})$, where σ^2 is the irreducible Poisson variance; at an intercept of −1 this variance accounted for 88–96% of the squared error against the observed response and so compressed all differences between models. Both are reported in Table 1, but only the error against the true mean is informative about how well the models estimate the conditional mean.

MAE was the clear exception. Its McFadden R^2 was negative (−0.098), indicating predictions worse than an intercept-only baseline, and it gave the highest RMSE against the true mean (0.295). Its Spearman correlation, however, was 0.586, indistinguishable from MSE and Gaussian. MAE therefore ranked observations as well as the other losses while placing them on the wrong scale. This is consistent with its target functional: MAE is minimised by the conditional median rather than the conditional mean, and at a mean count of approximately 0.37 the median is zero across most of the predictor space. MAE predicted below zero for 13.4% of test observations, and at least one negative prediction occurred in 89.6% of iterations, compared with 3.1% of observations for MSE, 2.3% for Gaussian and none for the two count losses. As described in the Methods, such predictions are clamped before the Poisson log-likelihood is

evaluated, which penalises them heavily; the magnitude of the negative R^2 therefore depends on the clamping bound, while its sign does not.

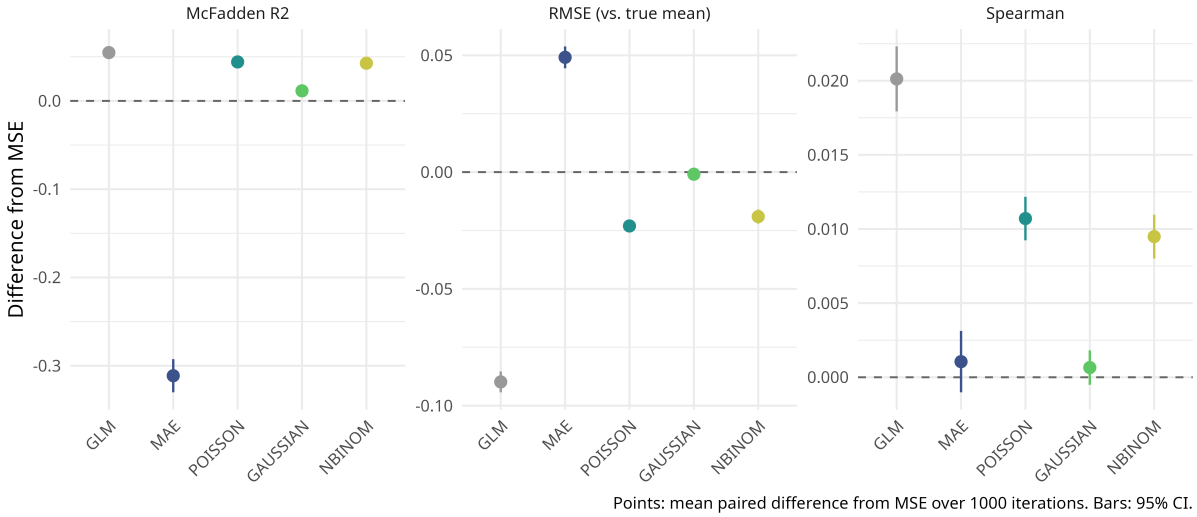


Figure 1. *Predictive accuracy of five DNN loss functions and the correctly specified GLM on Poisson-distributed data, expressed as paired differences from the MSE loss over 1000 iterations. Points show the mean difference, bars the 95% confidence interval; the dashed line marks no difference from MSE. Differences are paired within iteration, since all models were fitted to the same simulated dataset. For McFadden's R^2 and Spearman correlation, higher values are better and positive differences indicate an improvement on MSE; for RMSE against the true mean, lower is better and negative differences indicate an improvement. Note the differing vertical scales between panels. Absolute values are given in Table 1.*

Table 1. *Predictive accuracy of five DNN loss functions and the correctly specified GLM on Poisson-distributed data, averaged over 1000 iterations, with Monte Carlo standard errors in parentheses. McFadden's R^2 compares predictions with an intercept-only baseline under a Poisson likelihood and is not bounded below by zero out of sample. RMSE is reported against the true conditional mean and against the observed response; the former isolates estimation error from sampling noise. Spearman correlation assesses the predicted ordering independently of scale. All metrics are computed on held-out test data.*

Loss function	McFadden R2	RMSE (vs. true mean)	RMSE (vs. observed)	Spearman
GLM	0.268 (0.003)	0.156 (0.002)	0.832 (0.005)	0.605 (0.003)
MSE	0.213 (0.005)	0.246 (0.002)	0.853 (0.005)	0.585 (0.003)
MAE	-0.098 (0.012)	0.295 (0.002)	0.866 (0.005)	0.586 (0.003)
POISSON	0.257 (0.003)	0.223 (0.002)	0.846 (0.005)	0.596 (0.003)
GAUSSIAN	0.225 (0.004)	0.245 (0.002)	0.853 (0.005)	0.586 (0.003)
NBINOM	0.256 (0.003)	0.227 (0.002)	0.847 (0.005)	0.595 (0.003)

Effect estimation showed a different pattern (Figure 2). The GLM recovered all three effects essentially without bias (bias +0.006, −0.003 and +0.003 for Environment1 to Environment3). Every network underestimated the strongest effect, from −0.172 for MSE to −0.312 for MAE, corresponding to shortfalls of 13% to 23% relative to the true value of 1.367. Bias decreased

with effect size: partial η^2 for the loss function was 0.66 for Environment1, 0.10 for Environment2 and 0.005 for Environment3. This gradient is consistent with the weight decay applied during training, since L2 shrinkage is proportional to weight magnitude and therefore removes most from the largest effect.

The ordering of the losses differed between the two parts of the analysis. By RMSE against the true mean the ranking was Poisson, negative binomial, Gaussian, MSE, MAE; by absolute bias on the strongest effect it was MSE, Gaussian, Poisson, negative binomial, MAE. The loss that estimated the conditional mean most accurately was therefore not the one that recovered the effect most accurately, and the negative binomial loss in particular combined the second-best predictive accuracy with the second-largest bias. Matching the loss to the data-generating distribution improved prediction but did not improve effect recovery.

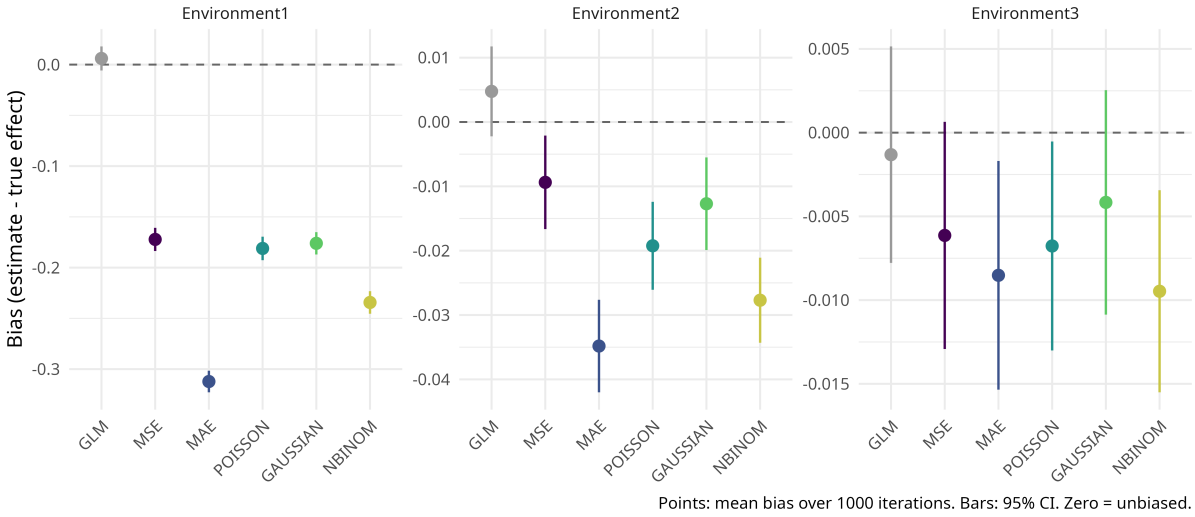


Figure 2. *Bias of estimated effects for five DNN loss functions and the correctly specified GLM on Poisson-distributed data, over 1000 iterations. Points show the mean difference between the estimated average conditional effect and the true average effect implied by the data-generating process for that iteration; bars give the 95% confidence interval. The dashed line marks zero bias. Note the differing vertical scales between panels.*

Coverage followed bias closely (Table 2). The GLM achieved 0.920 to 0.934, slightly below nominal; as noted in the Methods, a normal quantile applied to 20 bootstrap replicates implies approximately 0.935 even for correctly sized intervals, so this shortfall is largely a property of the interval construction rather than of the GLM. For the networks, coverage for Environment1 fell well below nominal, from 0.844 for the Gaussian loss to 0.701 for MAE, in the same order as bias. For Environment2 and Environment3, where bias was negligible, coverage was at or slightly above nominal (0.948–0.986). Coverage therefore tracked the displacement of the point estimates rather than any underestimation of their uncertainty.

Table 2. *Coverage and bias of effect estimates on Poisson-distributed data, over 1000 iterations. Coverage is the proportion of iterations in which estimate ± 1.96 SE contained the true effect for that iteration; bias is the mean difference between estimate and true effect. Monte Carlo standard errors in parentheses. Nominal coverage is* 0.95.

Loss function	Predictor	Coverage	Bias
MSE	Environment1	0.834 (0.012)	-0.172 (0.006)
MAE	Environment1	0.701 (0.014)	-0.312 (0.005)
POISSON	Environment1	0.822 (0.012)	-0.181 (0.006)
GAUSSIAN	Environment1	0.844 (0.011)	-0.176 (0.006)
NBINOM	Environment1	0.765 (0.013)	-0.234 (0.006)
GLM	Environment1	0.929 (0.008)	+0.006 (0.006)
MSE	Environment2	0.971 (0.005)	-0.009 (0.004)
MAE	Environment2	0.958 (0.006)	-0.035 (0.004)
POISSON	Environment2	0.952 (0.007)	-0.019 (0.003)
GAUSSIAN	Environment2	0.973 (0.005)	-0.013 (0.004)
NBINOM	Environment2	0.948 (0.007)	-0.028 (0.003)
GLM	Environment2	0.934 (0.008)	+0.005 (0.004)
MSE	Environment3	0.986 (0.004)	-0.006 (0.003)
MAE	Environment3	0.977 (0.005)	-0.009 (0.003)
POISSON	Environment3	0.970 (0.005)	-0.007 (0.003)
GAUSSIAN	Environment3	0.980 (0.004)	-0.004 (0.003)
NBINOM	Environment3	0.964 (0.006)	-0.009 (0.003)
GLM	Environment3	0.920 (0.009)	-0.001 (0.003)

Taken together, matching the loss function to the data-generating distribution improved estimation of the conditional mean by about 9% but did not improve the recovery of effects, while using a loss whose target functional differs from the mean — MAE — degraded both substantially.

Gaussian-distributed data

The first Gaussian runs, at an intercept of −1, produced usable results only for MSE, MAE and Gaussian: the Poisson and negative binomial losses failed to converge in every iteration, as roughly two thirds of simulated responses were negative and lie outside the support of both distributions. The scenario was therefore rerun at an intercept of 6, the lowest value producing reliably positive responses, at which all five losses converged.

No network matched the correctly specified GLM, but the ordering among the networks was not the one the Gaussian scenario was intended to show. Against the true mean the negative binomial loss gave the lowest error (0.213), followed by Poisson (0.235), Gaussian (0.243), MAE (0.255) and MSE (0.267), against 0.136 for the GLM (Table 3). Relative to the MSE loss this is a reduction of 20% for the negative binomial loss and 12% for Poisson, and all five intervals excluded zero (Figure 3). The same ordering held for R^2 (0.538 for MSE rising to 0.550 for negative binomial, against 0.562 for the GLM) and for Spearman correlation (0.741 to 0.749, against 0.753). The two count losses therefore held a small advantage even on Gaussian data, where they are misspecified. One possible explanation is that both overstate the variance here: at a mean of 6 and a residual standard deviation of 1 the data are strongly underdispersed relative to a Poisson, whose variance would equal the mean, and a loss assuming more noise than the data contain down-weights the residuals and so acts as additional regularisation. This remains a conjecture, since the design cannot separate it from the effect of the log link, which the count losses also impose.

As in the Poisson scenario, the apparent size of these differences depended on the scale of evaluation. Against the observed response the six models spanned 1.005 to 1.032, a range of 2.7%, whereas against the true mean they spanned 0.136 to 0.267, a range of 96%. The irreducible residual variance implied by the two is 0.99, matching the simulated residual standard deviation of 1, and accounts for 93–98% of the squared error against the observed response. Measured against the observed response alone, the differences between losses would again appear negligible.

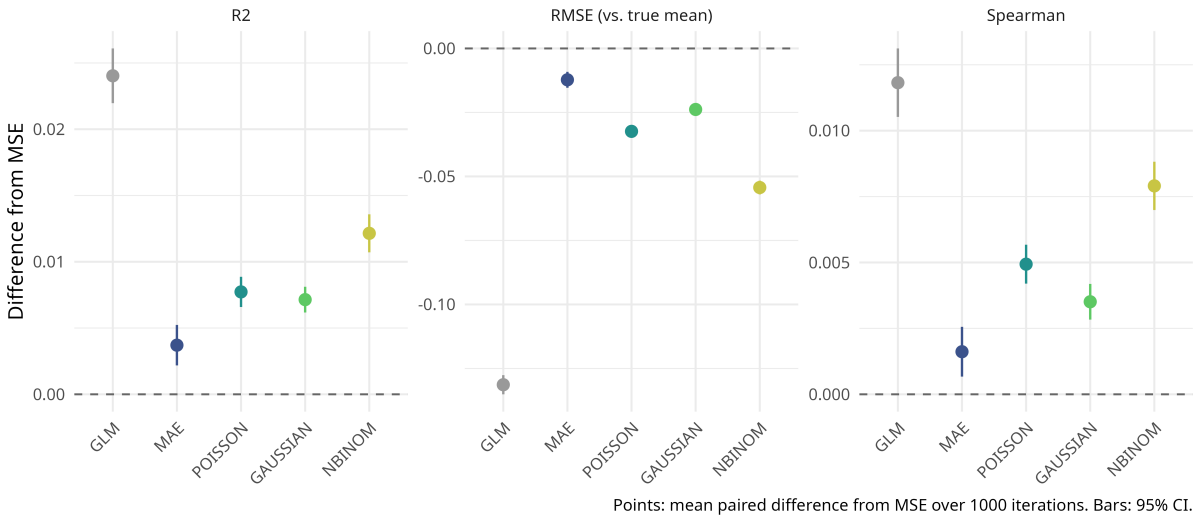


Figure 3. *Predictive accuracy of five DNN loss functions and the correctly specified GLM on Gaussian-distributed data, expressed as paired differences from the MSE loss over 1000 iterations. Points show the mean difference, bars the 95% confidence interval; the dashed line marks no difference from MSE. For R^2 and Spearman correlation, higher values are better and positive differences indicate an improvement on MSE; for RMSE against the true mean, lower is better and negative differences indicate an improvement. Note the differing vertical scales between panels. Absolute values are given in Table 3.*

Table 3. *Predictive accuracy of five DNN loss functions and the correctly specified GLM on Gaussian-distributed data, averaged over 1000 iterations, with Monte Carlo standard errors in parentheses. RMSE is reported against the true conditional mean and against the observed response; the former isolates estimation error from sampling noise. Spearman correlation assesses the predicted ordering independently of scale. All metrics are computed on held-out test data.*

Loss function	R2	RMSE (vs. true mean)	RMSE (vs. observed)	Spearman
GLM	0.562 (0.003)	0.136 (0.002)	1.005 (0.003)	0.753 (0.002)
MSE	0.538 (0.003)	0.267 (0.002)	1.032 (0.003)	0.741 (0.002)
MAE	0.542 (0.003)	0.255 (0.002)	1.028 (0.003)	0.742 (0.002)
POISSON	0.546 (0.003)	0.235 (0.002)	1.023 (0.003)	0.746 (0.002)
GAUSSIAN	0.545 (0.003)	0.243 (0.002)	1.024 (0.003)	0.744 (0.002)
NBINOM	0.550 (0.003)	0.213 (0.002)	1.018 (0.003)	0.749 (0.002)

Effect estimation showed the same ordering on a much smaller scale (Figure 4). All five networks underestimated the strongest effect, from −0.012 for the negative binomial loss to −0.058 for MAE, against −0.002 for the GLM; for Environment2 and Environment3 all biases lay within ±0.013 of zero. Relative to the true effect the largest network bias was 2.9%, against 12.6% to 22.8% under Poisson data, so the underestimation seen there was present but roughly seven times weaker rather than absent. Interval coverage showed the opposite pattern to the count scenario (Table 4): every network exceeded the nominal 95% for every predictor (0.959–0.991), so the intervals were wider than necessary rather than correctly sized, which is

consistent with the conservative standard errors cito reports under regularisation. The GLM fell slightly below nominal (0.919–0.937), close to the 0.935 implied by applying a normal quantile to 20 bootstrap replicates.

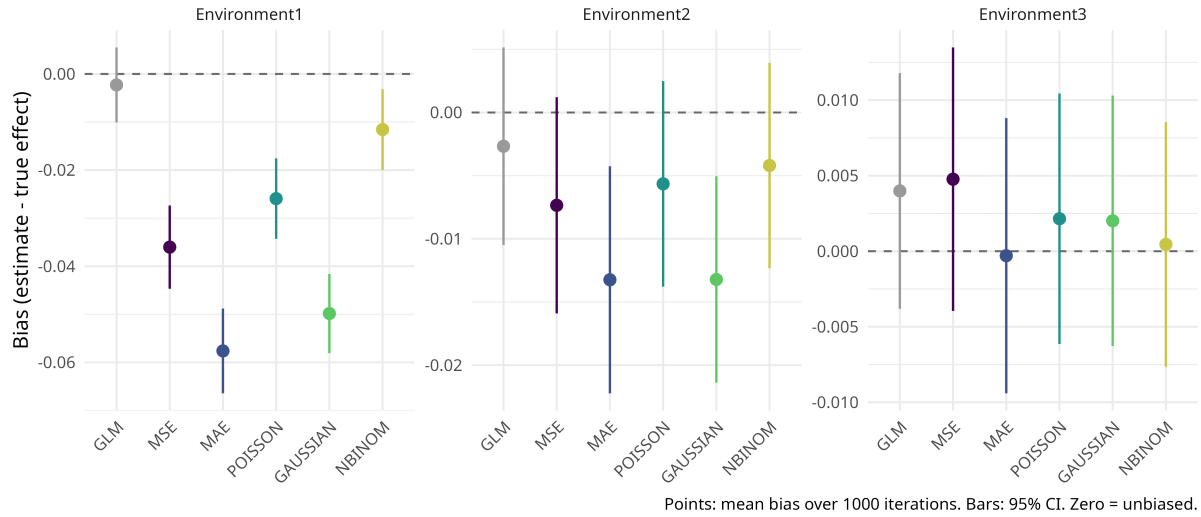


Figure 4. *Bias of estimated effects for five DNN loss functions and the correctly specified GLM on Gaussian-distributed data, over 1000 iterations. Points show the mean difference between the estimated average conditional effect and the true average effect implied by the data-generating process; bars give the 95% confidence interval. The dashed line marks zero bias. Note the differing vertical scales between panels.*

Table 4. *Coverage and bias of effect estimates for the three predictors, five DNN loss functions and the GLM baseline on Gaussian-distributed data, over 1000 iterations. Coverage is the proportion of iterations in which the estimate ± 1.96 SE contained the true effect; bias is the mean difference between estimate and true effect. Monte Carlo standard errors in parentheses. Nominal coverage is 0.95.*

Loss function	Predictor	Coverage	Bias
MSE	Environment1	0.975 (0.005)	-0.036 (0.004)
MAE	Environment1	0.974 (0.005)	-0.058 (0.004)
POISSON	Environment1	0.972 (0.005)	-0.026 (0.004)
GAUSSIAN	Environment1	0.959 (0.006)	-0.050 (0.004)
NBINOM	Environment1	0.991 (0.003)	-0.012 (0.004)
GLM	Environment1	0.937 (0.008)	-0.002 (0.004)
MSE	Environment2	0.975 (0.005)	-0.007 (0.004)
MAE	Environment2	0.971 (0.005)	-0.013 (0.005)
POISSON	Environment2	0.961 (0.006)	-0.006 (0.004)
GAUSSIAN	Environment2	0.961 (0.006)	-0.013 (0.004)
NBINOM	Environment2	0.965 (0.006)	-0.004 (0.004)
GLM	Environment2	0.919 (0.009)	-0.003 (0.004)
MSE	Environment3	0.973 (0.005)	+0.005 (0.004)
MAE	Environment3	0.982 (0.004)	-0.000 (0.005)
POISSON	Environment3	0.969 (0.005)	+0.002 (0.004)
GAUSSIAN	Environment3	0.974 (0.005)	+0.002 (0.004)
NBINOM	Environment3	0.966 (0.006)	+0.000 (0.004)
GLM	Environment3	0.927 (0.008)	+0.004 (0.004)

4. Discussion

The expectation set out in the introduction held in one scenario and failed in the other. On Poisson data the two count losses did perform best, reducing RMSE against the true conditional mean by 9% for the Poisson loss (95% CI [−0.026, −0.021]) and 8% for the negative binomial loss relative to the MSE default, with the matching Gaussian loss indistinguishable from MSE. On Gaussian data, where all five losses were expected to perform similarly, the two count losses again came out ahead, by 12% and 20%, while the nominally correct Gaussian loss did not differ from MSE at all (−0.001, 95% CI [−0.003, +0.001], p = 0.45). A matching loss therefore helped where it was expected to, but a mismatched loss helped as well where it was not. One possible explanation for the Gaussian result is that both count losses overstate

the variance of strongly underdispersed data and so down-weight the residuals, acting as additional regularisation; this remains a conjecture, since the design confounds the variance assumption with the log link that the count losses also impose, and separating the two would require fitting a loss with one component but not the other.

MAE on Poisson-distributed data was the one clear failure. Its negative McFadden R^2 indicates predictions worse than an intercept-only baseline, and the mechanism follows from what the loss optimises. Absolute error is minimised by the conditional median rather than the mean (Gneiting, 2011), and at an intercept of −1 the Poisson mean lies below 0.4, so the median count is zero across much of the predictor space. MAE predicted below zero for 13.4% of test observations against 3.1% for MSE and 2.3% for the Gaussian loss, and none for the count losses (Table S1). Because the Poisson log-likelihood is undefined at a non-positive rate, these predictions were clamped before evaluation, which penalises them heavily and produces the negative R^2. MAE's Spearman correlation, however, was indistinguishable from the other losses: the ordering of observations was recovered correctly and only the scale was wrong.

This failure has a direct precedent in the ecological literature. O'Hara & Kotze (2010) reported that linear models applied to log-transformed counts were severely biased relative to GLMs, and Morrissey & Ruxton (2020) showed that most of that apparent failure arose because the two analyses estimated different quantities: the linear model estimated $E[\log(y + 1)]$ while the GLM estimated $\log(E[y])$. Once both were expressed on the same scale, a linear model fitted to untransformed counts performed indistinguishably from the correctly specified GLM. The MAE result here is the same error in a different setting — a method consistent for the median scored as though it estimated the mean — and the near-equivalence of the MSE and Gaussian losses with the count losses on most metrics mirrors their finding that the distributional assumption matters less than ecological practice assumes, provided the quantity being estimated is the same.

The loss that predicted best was not the loss that estimated effects best. By RMSE against the true mean the ranking on Poisson data was Poisson, negative binomial, Gaussian, MSE, MAE; by absolute bias on the strongest effect it was MSE, Gaussian, Poisson, negative binomial, MAE. The negative binomial loss combined the second-best predictive accuracy with the second-largest bias, and every network underestimated the strongest effect by 13% to 23% while the GLM was unbiased. This dissociation may be genuine, or it may follow from the design: weight decay and early stopping were chosen to control overfitting, which is an accur-

acy criterion, and Pichler & Hartig (2023) note that hyperparameters optimised for predictive accuracy need not be optimal for inference. Either way, the result does not support extending a loss-matching rule from prediction to inference.

A second finding concerns how performance is measured. Against the observed response the six models spanned 0.832 to 0.866 on Poisson data, a range of 4%, whereas against the true conditional mean they spanned 0.156 to 0.295, a range of 89%. The irreducible sampling variance accounted for 88–96% of the squared error against the observed response and so compressed every difference between models. This matters for interpreting applied work: Montesinos-López et al. (2021) found that a Poisson-loss network did not outperform an MSE-loss network on count data, but assessed predictions against observed counts, which is the comparison in which an efficiency difference of the size found here would be almost invisible. Their companion study, in which generalised Poisson regression outperformed the corresponding networks (Montesinos-López et al., 2020), agrees with the GLM result reported here. Practitioner guidance shows the same blind spot from the other direction: it recommends MSE or MAE and distinguishes them almost entirely by robustness to outliers (Elharrouss et al., 2025; Lathuilière et al., 2020; Brownlee, n.d.; IBM, n.d.), while count-specific losses appear only as special cases, justified where at all by the link preventing negative predictions rather than by the variance assumption.

Three practical points follow. Defaulting to a loss that matches the response distribution is defensible for count data where predictive accuracy is the goal, since the count losses were ahead on every metric in both scenarios, though the gains are modest. For inference, nothing in these results supports such a rule, and the question of whether it should be advised against must be left to studies that tune for calibration rather than for accuracy. And defaulting to MAE carries a specific risk that current guidance does not mention: a loss targeting the wrong functional of the conditional distribution can fail outright on skewed count data, and the divergence between mean and median is largest exactly where counts are sparse.

The correctly specified GLM outperformed every network in both scenarios, by 43% on RMSE against the true mean under Poisson data and 57% under Gaussian data. This is expected rather than informative: specifying the family hands the GLM both the functional form and the link, while the networks must recover the form from the data, and the design favours the GLM throughout with a small sample, small effects, no interactions, no nonlinearity, no overdispersion and no random effects (Jansen et al., 2020; Wisz et al., 2008). These results therefore

say nothing about the case in which the parametric form is wrong, which is the case in which a network would be used at all. The much smaller gap against the observed response — 2.5% for MSE against the GLM — is a better guide to what the difference would look like on real data, where the true mean is unavailable.

Several limitations and extensions follow from the design. Neither architecture nor hyperparameters were tuned, and the learning rate was held constant across both distributions and all five losses, so some differences may reflect settings that suit one loss better than another. Regularisation is a more specific concern: weight decay and early stopping both shrink weights toward zero, which depresses the largest effect most in absolute terms, and partial η^2 for the loss function fell from 0.66 for the strongest effect to 0.005 for the weakest, consistent with that mechanism. A preliminary rerun without either reduced bias for the identity-link losses but reversed its sign for the two log-link losses, and widened the bootstrap intervals until coverage reached 1.000 and became uninformative; this suggests a bias–variance trade-off rather than a straightforward improvement, and the sign reversal is unexplained. On the data-generating side, introducing overdispersion or random-effect variance would move the simulated data closer to ecological datasets, and zero-inflation deserves separate attention, since at the intercept used here roughly two thirds of responses are zeros. Finally, a further extension would return to the question that motivated this study: inspecting simulated quantile residuals from the fitted networks directly, rather than relying on aggregate accuracy metrics, to establish whether a matched loss produces residuals that are demonstrably better behaved and whether that signal could guide loss selection in practice.

In conclusion, matching the loss function to the data-generating distribution produced a real but modest improvement in estimating the conditional mean, and no improvement in recovering effects. The size of that improvement depends on how performance is measured: against observed responses it is almost invisible, and against the true mean it is of the order of 10%. The clearest practical result is negative rather than positive — mean absolute error, which standard guidance presents as a routine alternative to squared error, fails on sparse count data because it estimates the median rather than the mean. Choosing a loss means choosing a distributional assumption whether or not that choice is made deliberately, and the component of it that matters most in these results is not which distribution is named but which quantity the loss is consistent for.

AI-Policy

I used Anthropics' Claude Opus 5 for debugging and cleaning code, aiding in literature search and reviewing this text. Study design, interpretation and final text were not delegated to AI.

Reproducibility

For reproducibility, the L'Ecuyer-CMRG random number generator was used so that parallel workers receive reproducible streams, with seeds set via set.seed() and, because torch maintains its own generator for weight initialisation, torch_manual_seed() within each worker. All analyses used R 4.5.3 with cito 1.2, DHARMa 0.5.0 and marginaleffects 1.0.0. Other packages I used were pbmcapply 1.5.1, stringr 1.6.0, dplyr 1.2.1, ggplot2 4.0.3, tidyr 1.3.2 and torch 0.17.0

Code and Data are available under https://github.com/YannikGit/ProjektPraktikum.

References

Amesöder, C., Hartig, F., & Pichler, M. (2024). "cito": An R package for training neural networks using "torch." *Ecography*, *2024*(6), e07143. https://doi.org/10.1111/ecog.07143

Brownlee, J. (n.d.). *How to Choose Loss Functions When Training Deep Learning Neural Networks*. Machine Learning Mastery. Retrieved October 12, 2026, from https://machinelearningmastery.com/how-to-choose-loss-functions-when-training-deep-learning-neural-networks/

Elharrouss, O., Mahmood, Y., Bechqito, Y., Serhani, M. A., Badidi, E., Riffi, J., & Tairi, H. (2025). *Loss Functions in Deep Learning: A Comprehensive Review*. arXiv. https://doi.org/10.48550/arXiv.2504.04242

Fisher, R. A. (1922). On the mathematical foundations of theoretical statistics. *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character*, *222*(594–604), 309–368. https://doi.org/10.1098/rsta.1922.0009

Gneiting, T. (2011a). Making and Evaluating Point Forecasts. *Journal of the American Statistical Association*, *106*(494), 746–762. https://doi.org/10.1198/jasa.2011.r10138

Gneiting, T. (2011b). Making and Evaluating Point Forecasts. *Journal of the American Statistical Association*, *106*(494), 746–762. https://doi.org/10.1198/jasa.2011.r10138

Gneiting, T., & Raftery, A. E. (2007). Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, *102*(477), 359–378. https://doi.org/10.1198/016214506000001437

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. https://www.deeplearningbook.org/

IBM. (n.d.). *What is a loss function?* Retrieved October 12, 2026, from https://www.ibm.com/think/topics/loss-function

Ives, A. R. (2015). For testing the significance of regression coefficients, go ahead and log-transform count data. *Methods in Ecology and Evolution*, *6*(7), 828–835. https://doi.org/10.1111/2041-210X.12386

Jansen, J., Woolley, S. N. C., & Dunstan, P. K. (2020). Effects of sample size and network depth on a deep learning approach to species distribution modeling. *Ecological Informatics*, *60*, 101137. https://doi.org/10.1016/j.ecoinf.2020.101137

Lathuilière, S., Mesejo, P., Alameda-Pineda, X., & Horaud, R. (2020). A Comprehensive Analysis of Deep Regression. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *42*(9), 2065–2081. https://doi.org/10.1109/TPAMI.2019.2910523

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436–444. https://doi.org/10.1038/nature14539

McCullagh, P., & Nelder, J. A. (1989). *Generalized Linear Models* (2nd ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9780203753736

Montesinos-López, O. A., Montesinos-López, J. C., Salazar, E., Barron, J. A., Montesinos-López, A., Buenrostro-Mariscal, R., & Crossa, J. (2021). Application of a Poisson deep neural network model for the prediction of count data in genome-based prediction. *The Plant Genome*, *14*(3), e20118. https://doi.org/10.1002/tpg2.20118

Montesinos-López, O. A., Montesinos-López, J. C., Singh, P., Lozano-Ramirez, N., Barrón-López, A., Montesinos-López, A., & Crossa, J. (2020). A Multivariate Poisson Deep Learning

Model for Genomic Prediction of Count Data. *G3 Genes|Genomes|Genetics*, *10*(11), 4177–4190. https://doi.org/10.1534/g3.120.401631

Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, *38*(11), 2074–2102. https://doi.org/10.1002/sim.8086

Morrissey, M. B., & Ruxton, G. D. (2020). Revisiting advice on the analysis of count data. *Methods in Ecology and Evolution*, *11*(9), 1133–1140. https://doi.org/10.1111/2041-210X.13372

O'Hara, R. B., & Kotze, D. J. (2010). Do not log-transform count data. *Methods in Ecology and Evolution*, *1*(2), 118–122. https://doi.org/10.1111/j.2041-210X.2010.00021.x

Oliveira, N. L., Lei, J., & Tibshirani, R. J. (2025). Unbiased test error estimation in the Poisson means problem via coupled bootstrap techniques. *Electronic Journal of Statistics*, *19*(1), 361–396. https://doi.org/10.1214/24-EJS2336

Pichler, M., & Hartig, F. (2023a). Machine learning and deep learning—A review for ecologists. *Methods in Ecology and Evolution*, *14*(4), 994–1016. https://doi.org/10.1111/2041-210X.14061

Pichler, M., & Hartig, F. (2023b). Machine learning and deep learning—A review for ecologists. *Methods in Ecology and Evolution*, *14*(4), 994–1016. https://doi.org/10.1111/2041-210X.14061

Terven, J., Cordova-Esparza, D.-M., Romero-González, J.-A., Ramírez-Pedraza, A., & Chávez-Urbiola, E. A. (2025). A comprehensive survey of loss functions and metrics in deep learning. *Artificial Intelligence Review*, *58*(7), 195. https://doi.org/10.1007/s10462-025-11198-7

Ver Hoef, J. M., & Boveng, P. L. (2007). Quasi-Poisson vs. Negative binomial regression: How should we model overdispersed count data? *Ecology*, *88*(11), 2766–2772. https://doi.org/10.1890/07-0043.1

Warton, D. I., Lyons, M., Stoklosa, J., & Ives, A. R. (2016). Three points to consider when choosing a LM or GLM test for count data. *Methods in Ecology and Evolution*, *7*(8), 882–890. https://doi.org/10.1111/2041-210X.12552

Wedderburn, R. W. M. (1974). Quasi-likelihood functions, generalized linear models, and the Gauss—Newton method. *Biometrika*, *61*(3), 439–447. https://doi.org/10.1093/biomet/61.3.439

White, H. (1982). Maximum Likelihood Estimation of Misspecified Models. *Econometrica*, *50*(1), 1–25. https://doi.org/10.2307/1912526

Wisz, M. S., Hijmans, R. J., Li, J., Peterson, A. T., Graham, C. H., & Guisan, A. (2008). Effects of sample size on the performance of species distribution models. *Diversity and Distributions*, *14*(5), 763–773. https://doi.org/10.1111/j.1472-4642.2008.00482.x

Appendix

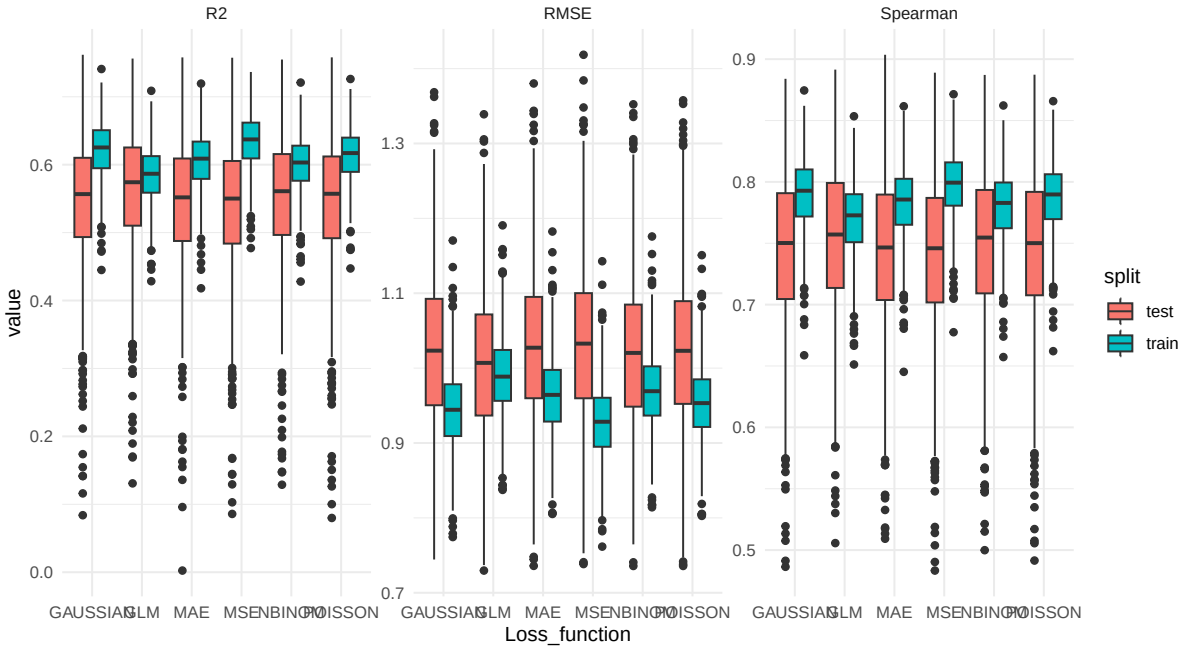


Figure S1. *Train–test comparison of accuracy metrics on Gaussian-distributed data, over 1000 iterations. Boxplots show the distribution of R², RMSE and Spearman correlation across iterations, calculated separately on training and held-out test data for each model. Substantially better performance on training than on test data would indicate overfitting. All models show a small and comparable train–test gap, with no loss function diverging from the others; the GLM shows the smallest gap, as expected for a four-parameter model.*

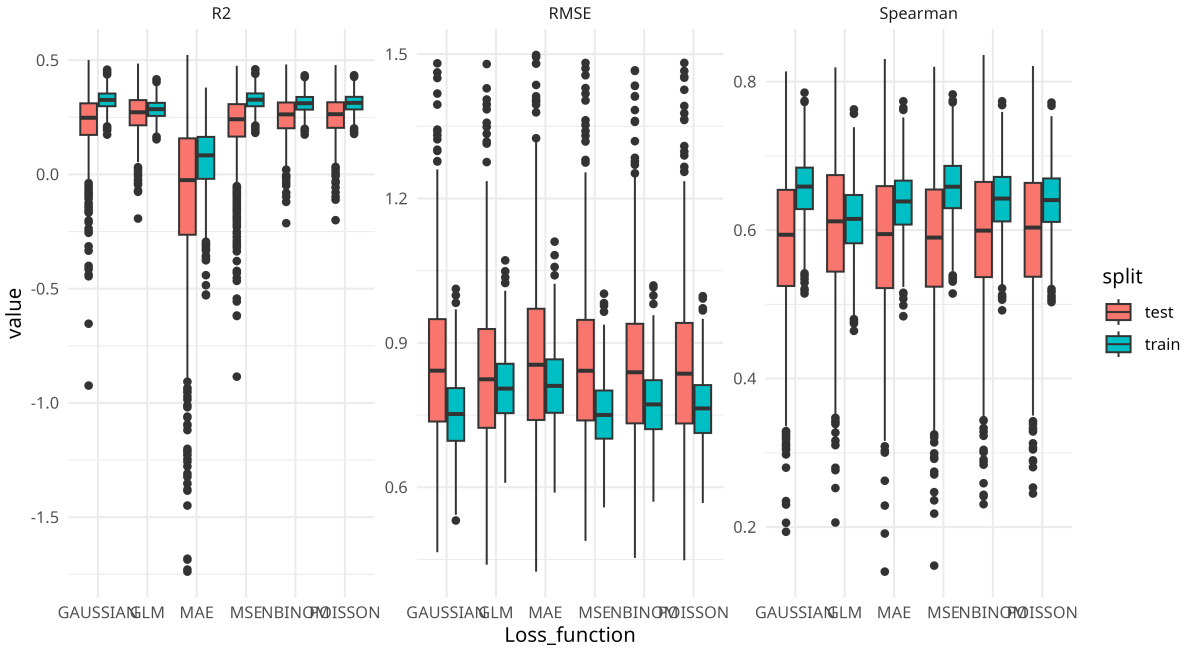


Figure S2. *Train–test comparison of accuracy metrics on Poisson-distributed data, over 1000 iterations. Boxplots show the distribution of R², RMSE and Spearman correlation across iterations, calculated separately on training and held-out test data for each model. Substantially better performance on training than on test data would indicate overfitting. All models show a small and comparable train–test gap, with only MAE diverging from the others on R², which is discussed at length in the main text; the GLM shows the smallest gap, as expected for a four-parameter model.*

Table S1. *Negative predictions by the GLM baseline and five DNN loss functions on Poisson-distributed data, over 1000 iterations with 50 test observations each. Columns give the mean number of negative predictions per iteration, the same figure as a percentage of test observations, the percentage of iterations containing at least one negative prediction, and the largest number observed in any single iteration. Negative values are impossible for count data and indicate predictions outside the support of the response.*

Loss_function	mean_neg	pct_neg	iter_any	max_neg
MSE	1.5	3.1	57.4	14
MAE	6.7	13.4	89.6	27
POISSON	0.0	0.0	0.0	0
GAUSSIAN	1.1	2.3	48.9	10
NBINOM	0.0	0.0	0.0	0
GLM	0.0	0.0	0.0	0